

```

pi@pi:~$ sudo systemctl status tor@default.service
● tor@default.service - Anonymizing overlay network for TCP
   Loaded: loaded (/lib/systemd/system/tor@default.service; enabled-runtime; vendor preset: enabled)
   Active: active (running) since Tue 2021-06-08 18:24:09 BST; 2 days ago
     Process: 9140 ExecStartPre=/usr/bin/install -Z -m 02755 -o debian-tor -g debian-tor -d /run/tor (code=exited, status=0/SUCCESS)
     Process: 9141 ExecStartPre=/usr/bin/tor --defaults-torrc /usr/share/tor/tor-service-defaults-torrc -f /etc/tor/torrc --RunAsDaemon 0 --va
Main PID: 9143 (tor)
    Tasks: 1 (limit: 2200)
   Memory: 29.1M
    CGroup: /system.slice/system-tor.slice/tor@default.service
            └─9143 /usr/bin/tor --defaults-torrc /usr/share/tor/tor-service-defaults-torrc -f /etc/tor/torrc --RunAsDaemon 0

Jun 10 12:24:09 pi Tor[9143]: Heartbeat: Tor's uptime is 1 day 18:00 hours, with 0 circuits open. I've sent 331 kB and received 1.59 MB.
Jun 10 12:24:09 pi Tor[9143]: Average packaged cell fullness: 70.435%. TLS write overhead: 9%
Jun 10 18:24:09 pi Tor[9143]: Heartbeat: Tor's uptime is 2 days 0:00 hours, with 0 circuits open. I've sent 345 kB and received 1.78 MB.
Jun 10 18:24:09 pi Tor[9143]: Average packaged cell fullness: 70.535%. TLS write overhead: 9%
Jun 11 00:24:09 pi Tor[9143]: Heartbeat: Tor's uptime is 2 days 6:00 hours, with 0 circuits open. I've sent 364 kB and received 2.01 MB.
Jun 11 00:24:09 pi Tor[9143]: Average packaged cell fullness: 70.678%. TLS write overhead: 9%
Jun 11 06:24:09 pi Tor[9143]: Heartbeat: Tor's uptime is 2 days 12:00 hours, with 0 circuits open. I've sent 376 kB and received 2.21 MB.
Jun 11 06:24:09 pi Tor[9143]: Average packaged cell fullness: 70.734%. TLS write overhead: 9%
Warning: Journal has been rotated since unit was started. Log output is incomplete or unavailable.
pi@pi:~$

```

Figure 1: Output showing Tor server status

If you prefer any other distros, you will need to modify the commands accordingly. Open the terminal and update the system with the following commands:

```

$ sudo apt-get update
$ sudo apt-get upgrade

```

Now we have to install the Tor server, as follows:

```

$ sudo apt-get install tor

```

Once the installation is successfully completed, we need to test if the service is running:

```

$ sudo systemctl status tor@default.
service

```

The output of the above command will be as shown in Figure 1.

Configuration

Since the server is up and running now, it's time for some tweaks to make the service accessible from other machines. To start with, we have to make some changes in the file `/etc/tor/torrc`. But before making any change, create a backup of the original file. Open the file in any text editor:

```

$ sudo cp /etc/tor/torrc /etc/tor/
torrc.backup
$ sudo nano /etc/tor/torrc

```

To make the service available on Port 9050 and accept connections from the LAN, include the following lines at the end of the file, assuming the IP address of the machine is 192.168.1.100. (It may be different for you as it depends on your network environment.)

```

SocksPort 192.168.1.100:9050
SocksPolicy accept 192.168.1.0/24
RunAsDaemon 1
DataDirectory /var/lib/tor

```

Save and close the file using `ctl+o` and `ctl+x`. Restart the service to make the changes effective:

```

$ sudo systemctl restart tor@default.
service

```

Using the Tor server

Now that we are done with the configuration and setting up of the Tor proxy server, we need to configure our browsers in the other machines in order to access the service and be anonymous. To start with, we have to open the network settings. As an example, for the Mozilla Firefox browser, we can open *Settings*, and then click on *Network Settings* at the bottom. We will be

redirected to a page where we will see the *No Proxy* option has been selected by default. We need to change that to *Manual Proxy Configuration*. In the *SOCKS Host*, we have to put the IP address of the machine where we have installed the Tor server. In my case, it's 192.168.1.100. In the adjacent port field, we have to add the port of the Tor service, 9050.

You are all set now, and it's time to go anonymous and enjoy your own setup.

To test your work, visit www.dnsleaktest.com and check the IP address it shows, which should be the IP exposed to the target websites instead of your original IP.

Choose the proxy country

We will now be able route our traffic through a random IP provided by the Tor service. However, we may need to use the IP of a particular country to access some content, in which case we need to make a few more changes to the file `/etc/tor/torrc`, which we have already altered earlier above. A word of caution: before you use Tor, be aware of any limitations imposed on using it by your organisation or the country you live in. **END** 🐧

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